

# In your books

Write word equations for the reactions below

a. Lithium and fluorine react together to produce lithium fluoride



b. Magnesium and oxygen react together to produce magnesium oxide.



Complete Mid Topic  
assessment on Google  
classroom

Due Wednesday 7th  
February

# Conservation of mass and balanced equations



Queen  
Elizabeth  
School

# Learning objectives

- explain the law of conservation of mass
- explain why a multiplier appears as a subscript in a formula
- explain why a multiplier appears in equations before a formula.

Keywords

**balanced symbol equation**

**conservation of mass**

**products**

**reactants**

Just a minute

**Speak for a minute without hesitation, repetition or deviations  
about  
Global Warming.**

- What is global warming
- What are the effects of global warming
- How it can be changed
- Why is it harmful/useful?



When chemicals react, atoms are not made or destroyed, they are just rearranged.

As a result, the total mass is always conserved during a chemical reaction.

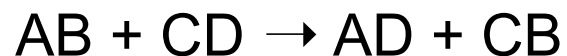
Because of this, chemical reactions can be represented by balanced symbol equations.

### *The law of conservation of mass*

The law of **conservation of mass** states that 'no atoms are lost or made during a chemical reaction'.

This means that the mass of the **products** equals the mass of the **reactants**.

In a chemical reaction, all the atoms in the reactants are *rearranged* into products. For example, for one type of reaction:



the mass of atoms at the start equals the mass of atoms at the finish. This is consistent with the law of 'conservation of mass'.

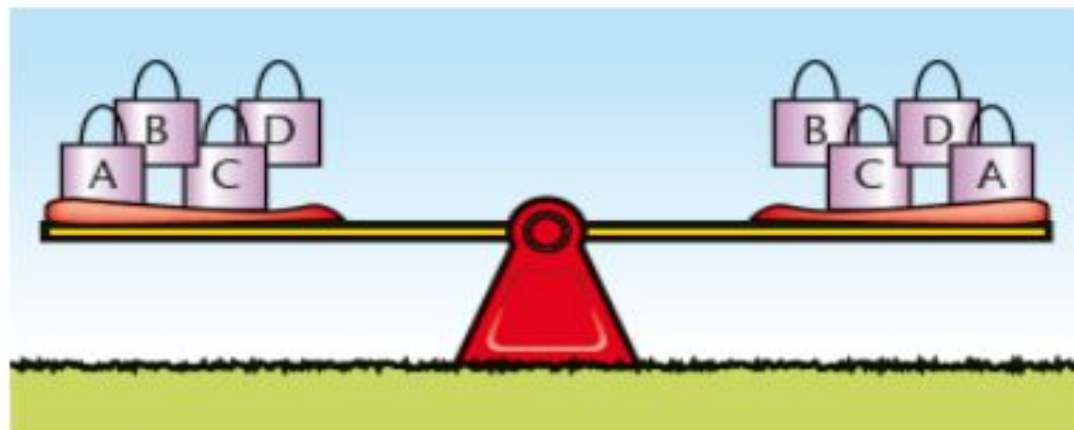


Figure 3.1 Masses of reactants and products are balanced.

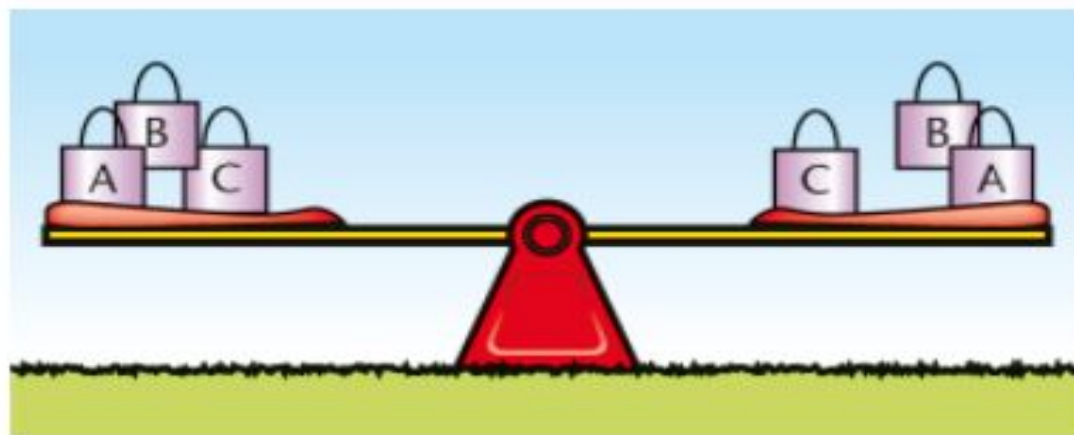


Figure 3.2 Numbers of atoms of the formulae are balanced.



Chemical equations are a simple way to describe how different substances are made in a reaction.

The equation tells us which substances have reacted in the reaction.

These are called the reactants and are found on the left-hand side of the equation.

The equation tells us which substances are produced in the reaction.

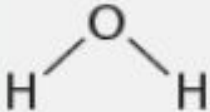
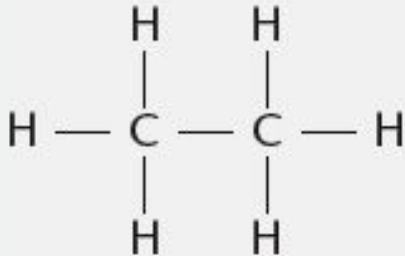
These are called the products and are found on the right-hand side of the equation

Sodium and chlorine react together to produce sodium chloride.



Many compounds contain more than one atom of an element in a formula. The number of atoms of the element is written as a subscript that is written after the element symbol.

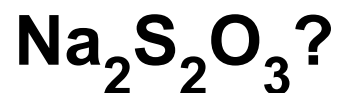
For example, these formulae have more than one atom of some of their elements. If there is no number after the element symbol it means 1 atom.

|                |                        |                                                                                      |
|----------------|------------------------|--------------------------------------------------------------------------------------|
| carbon dioxide | $\text{CO}_2$          | $\text{O}=\text{C}=\text{O}$                                                         |
| water          | $\text{H}_2\text{O}$   |   |
| ethane         | $\text{C}_2\text{H}_6$ |  |

1) Determine the number of hydrogen atoms in ethanoic acid



2) How many atoms of each element are in sodium thiosulfate



Answer- Na- 2

S- 2

O-3

3) How many atoms of each element are in aluminium sulfate



Answer-Al =2

S=3

O=12

How many atoms of each element are present in the compounds below?

a.  $\text{CO}_2$  1 atom of carbon 2 atoms of oxygen

b.  $\text{H}_2\text{SO}_4$  2 atoms of hydrogen 1 atom of sulfur 4 atoms of oxygen

c.  $\text{Mg}(\text{NO}_3)_2$  1 atom of magnesium 2 atoms of nitrogen 6 atoms of oxygen

Symbol equations give us more detail as they show the elements present in the reaction and also the number of each element.

To balance a symbol equation:



<https://youtube.com/watch?v=BRjwchL-AC8&feature=share>

## To balance a symbol equation:

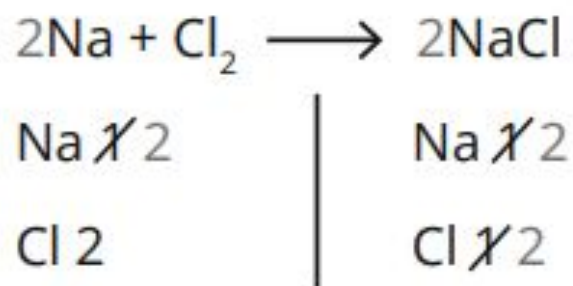
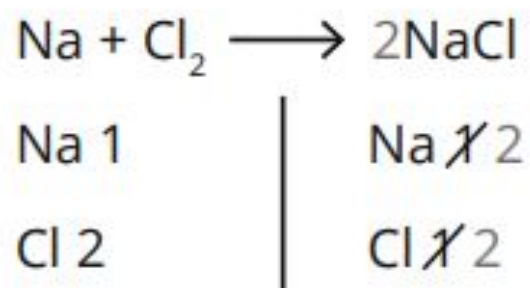
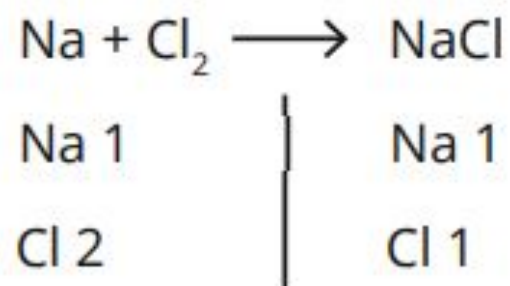
**Step 1** – Write the symbol equation.

**Step 2** – List the elements that are present on each side of the arrow.

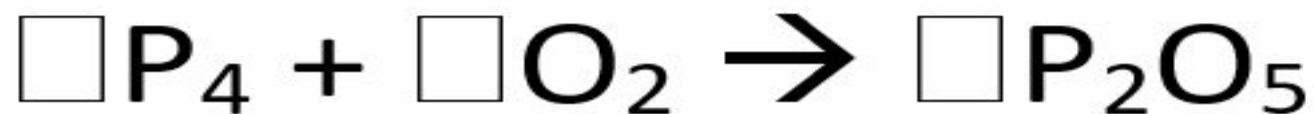
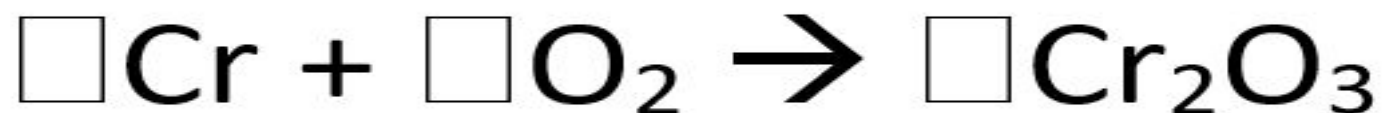
**Step 3** – Write down the number of each element present on each side.

**Step 4** – Identify which elements have different amounts on each side.

**Step 5** – Through trial and error, place numbers in front of the chemical formulae to balance the equation.



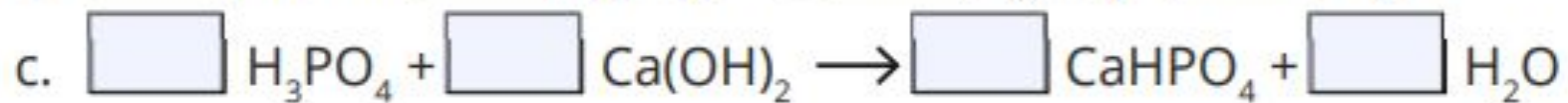
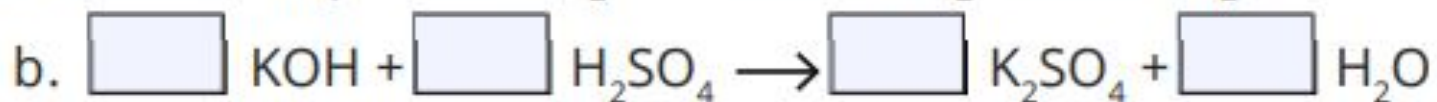
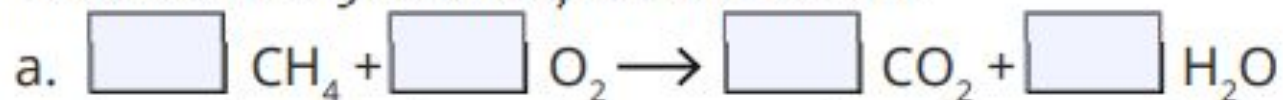
# Balance the equations



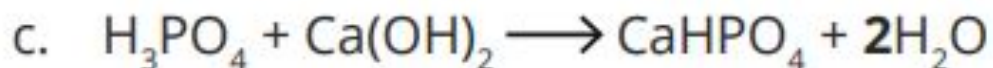
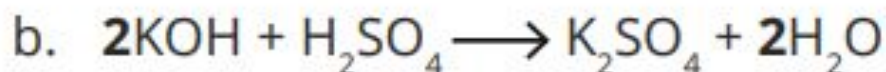


# Balance the equations

Balance the symbol equations below.



Balance the symbol equations below.





# Conservation of Mass Analogy

In a **closed system**, the **mass** in a chemical reaction will remain **constant**. In an **open system**, **changes in mass** can occur.

For example, if a gas is released and escapes from an open system, the total mass will look as if it has decreased. Similarly, if a gas is gained, the total mass will look as if it has increased. The total mass will remain the same only if the mass of the gas is accounted for.

**No atoms are created nor destroyed; they are just rearranged.**

However, in reality, it is not always possible to get the exact calculated product from a chemical reaction. Not all reactions react as expected. Some reversible reactions may not react fully.

In this closed system (the classroom), the mass in the reaction remains constant. As the system is closed, no children can leave or enter.



In this open system (the classroom), the mass in the reaction can change as children can leave or enter the classroom.

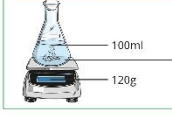


# Progress Check

**BEYOND SCIENCE**

## Conservation of Mass

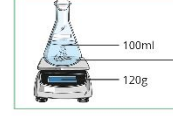
Gold

|                                                                                                                                |                                                                                            |                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|  <p>lemonade<br/>173g</p>                     |  <p>7</p> | Explain what happened to the mass of the lemonade after 24 hours.<br>_____<br>_____<br>_____ |
|  <p>100ml<br/>water<br/>salt<br/>110g</p>     |  <p>7</p> | Explain what happened to the mass of the salt solution.<br>_____<br>_____<br>_____           |
|  <p>100ml<br/>magnesium and acid<br/>120g</p> |                                                                                            | Explain what happened to the mass of the magnesium and acid.<br>_____<br>_____<br>_____      |

**BEYOND SCIENCE**

## Conservation of Mass

Silver

|                                                                                                                                  |                                                                                              |                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|  <p>lemonade<br/>173g</p>                     |  <p>7</p> | After 24 hours, the mass of the lemonade [increased, decreased, stayed the same].<br>Explain why.<br>_____<br>_____<br>_____ |
|  <p>100ml<br/>water<br/>salt<br/>110g</p>     |  <p>7</p> | The mass of the salt solution [increased, decreased, stayed the same].<br>Explain why.<br>_____<br>_____<br>_____            |
|  <p>100ml<br/>magnesium and acid<br/>120g</p> |                                                                                              | The mass of the magnesium and acid [increased, decreased, stayed the same].<br>Explain why.<br>_____<br>_____<br>_____       |

# Practical Investigation

Step 4 - Place the crucible on a lid on top of the following pieces of equipment and heat strongly with a blue flame for five minutes.

**crucible and lid, Bunsen burner, tripod, pipe clay triangle, heatproof mat, tongs, top pan balance, 2cm strip of magnesium.**

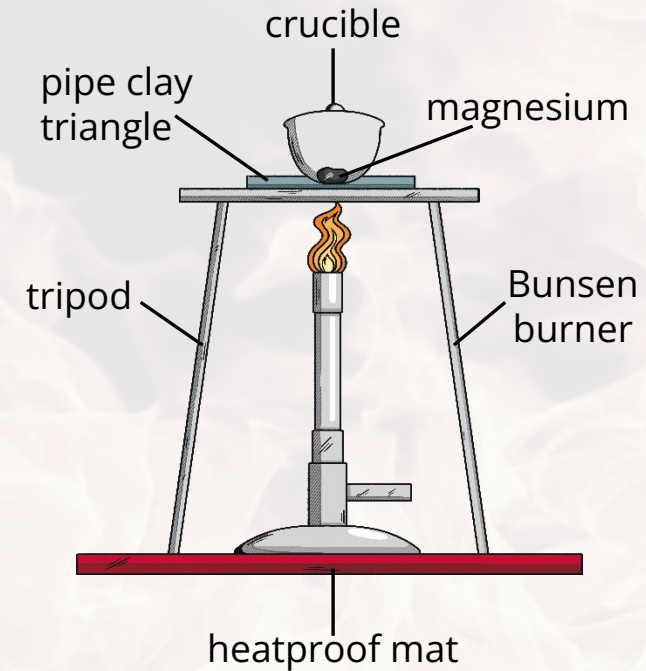
Step 1 - Once the magnesium begins to burn, the magnesium will glow a bright white. Using tongs, lift the lid for a moment so that you are able to observe the reaction taking place. Do not stare directly at the burning magnesium.

Step 2 - Place the crucible containing the magnesium onto the top pan balance and record the mass in the results table. Be sure to include the lid of the crucible in your measurement. Before taking the measurement ensure you tare the balance.

Step 3 - Measure the mass again.

Step 4 - Record the values in your table.

Step 5 - Place the Bunsen burner on top of a heatproof mat. Position a tripod over the Bunsen burner so that the barrel of the Bunsen burner is in the centre of triangle. Place the pipe clay triangle on top of the tripod.



# Conclusion and Evaluation

## Conclusion

What did you find out from the practical? How could you use your results to support your answer?

## Evaluation

If you were asked to complete the practical again, what would you do to improve it?